Zhang 2021 RCC — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

1 connected component at all θ; iLISI peak 3.12 @ θ=10; the axis recurs within patients (not a between-patient integration artifact).

Gate 2 — module geography → GO (non-branching)

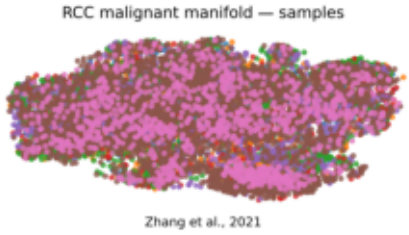
Non-branching: no Gast/pSom co-expression hub (joint co-high 4.9% vs 11.6% expected); pSom↔Crest inverse terminal gradient r=−0.80, Gast/pSom cross-repression r=−0.53. Single organized axis, no divergent arms.

Gate 3 — cross-repression → NA

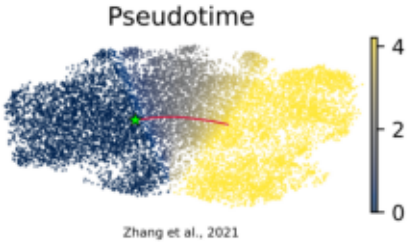
Not applicable — no bifurcation / hub to test for Gast↔pSom node anti-correlation.

STEP FIGURES

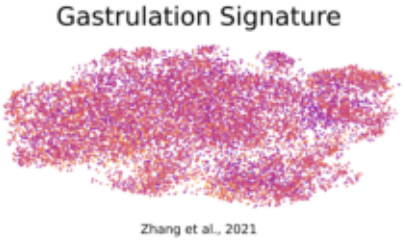
Manifold — samples



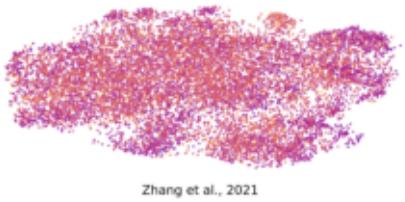
Pseudotime



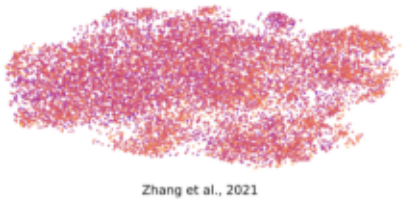
Gastrulation



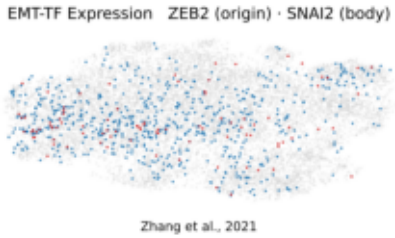
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Clear-cell + chromophobe renal cell carcinoma (Zhang et al., 2021), 10x — 11,410 malignant cells → 11,218 after per-sample threshold → 11,136 after island removal, across 7 samples (6 ccRCC + 1 chromophobe); the whole malignant manifold enters the trajectory (non-branching linear geometry).

Verdict.
The embryonic pattern reproduces as the non-branching case. A single continuous manifold carries an organized posterior-somitogenesis (origin) ↔ neural-crest (terminal) inverse axis (r=−0.80) with Gast/pSom cross-repression (r=−0.53), but no Gast/pSom co-expression hub and no divergent arms — so no bifurcation. Geometry LINEAR; Gate 3 NA. The axis is genuine intra-tumoral biology: it reproduces within every individual patient (mean pSom↔Crest r=−0.57), not a between-patient integration artifact. EMT programme is origin-loaded (all 6 expressed TFs decline from the origin).

Parameter choices

Min-cell threshold	369 (geom-mean of dominant gap 192→710, ratio 3.70); drops 1 sample	**Over-dispersed genes**	1,363 (scFates find_overdispersed); n_pcs=20 (scree knee)
Integration	Harmony on sample, θ=10 (iLISI peak 3.12; θ=20 re-segregates chromophobe)	**Island removal**	radius graph 6× median NN; keep largest comp (99.3%); 82 fringe cells
Geometry	LINEAR — ElPiGraph curve (Nodes 12, λ 2.0); 2 tips, 0 forks, fold-free (2.99)	**Root**	Gast/pSom-high, Crest-low curve tip (developmental origin; Crest is late)
Module means	Gast −0.034 / pSom +0.085 / Crest −0.060 (Crest terminal, no Gast arm)	**EMT-TFs**	ZEB2 (23.9%, origin) / SNAI2 (3.3%, body); all origin-loaded, 0 terminus factor